BOND RISK AND RETURN USING DURATION AND CONVEXITY

Learning Outcome

- calculate the percentage price change of a bond for a specified change in yield, given the bond's duration and convexity

We now show how to estimate the percentage price change of a bond for a specified yield change, given the bond's duration and convexity. As we will show, such an estimate is substantially closer to the bond's actual price change than if estimated using duration alone, allowing for more efficient risk management.

Continuing with our focus on BRWA's five-year, 3.2% (semiannual) coupon bond priced at par for settlement on 15 October 2025 and maturity on 15 October 2030, if its yield-to-maturity changes by ±100 bps from 3.20%, then the actual prices, using Excel's PRICE function, would be as follows:

PV_+ = PRICE(DATE(2025,10,15),DATE(2030,10,15),0.032,0.0420,100,2,0) = 95.53212.

PV_- = PRICE(DATE(2025,10,15),DATE(2030,10,15),0.032,0.0220,100,2,0) = 104.71035.

So, the actual price–yield relationship for this bond is as follows:

- 100 bp increase in yield (4.20%): Price = 95.53212; $\%\Delta PV^{Full} = -4.46788\%$.
- 100 bp decrease in yield (2.20%): Price = 104.71035; $\%\Delta PV^{Full} = 4.71035\%$.

The modified duration for this bond was previously found to be 4.58676, so a 100 bp increase (decrease) in yield-to-maturity results in $\%\Delta PV^{Full} \approx -4.58676\%$ (4.58676%).

We now show the enhanced precision and risk reduction achieved by adding the convexity adjustment to the duration estimate when determining the BRWA bond's percentage price changes. Using Equation 1 and the *AnnConvexity* metric, for ±100 bp changes in yield-to-maturity (from 3.20%), we have the following:

$$\%\Delta PV^{Full} \approx (-4.58676 \times 0.0100) + [\tfrac{1}{2} \times 24.23895 \times (0.0100)^2] = -4.46556\%.$$

$$\%\Delta PV^{Full} \approx (-4.58676 \times -0.0100) + [\tfrac{1}{2} \times 24.23895 \times (-0.0100)^2] = 4.70795\%.$$

Exhibit 4 compares the estimated percentage changes in BRWA's bond prices for ±100 bp changes in yield using duration alone and using duration plus convexity against the exact changes from Excel's PRICE function. It is clear that enhanced precision is achieved by adding the convexity adjustment, because the percentage price difference is less than ¼ of a basis point by adding convexity versus about 12 bps with duration alone. Note that this difference would be substantially greater for larger changes in yield or for longer-maturity or lower-coupon bonds.

Exhibit 4:

Using Duration and Convexity to Estimate Bond Price Changes

Rate Your Confidence
High
Medium
Low
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Change in Bond Yield	Actual $\%\Delta PV^{Full}$	$\%\Delta PV$ Using *ModDur*	Difference from Actual Change	$\%\Delta PV$ Using *ModDur* + Convexity	Difference from Actual Change
+ 100 bps	–4.46788	–4.58676	–0.11887	–4.46556	0.00232
– 100 bps	4.71035	4.58676	–0.12359	4.70795	–0.00239

Recall that money duration indicates the first-order effect on the full price of a bond in currency units given a change in yield-to-maturity. **Money convexity** (*MoneyCon*) captures the second-order effect in currency terms and is the annual convexity multiplied by the full price, as in Equation 3.

$$MoneyCon = AnnConvexity \times PV^{Full}. \quad (3)$$

Similar to estimating the percentage change in a bond's full price, MoneyDur and MoneyCon are combined to achieve a more accurate, thus less risky, estimate of the change in a bond's full price, as shown in Equation 4.

$$\Delta PV^{Full} \approx -(MoneyDur \times \Delta Yield) + \left[\frac{1}{2} \times MoneyCon \times (\Delta Yield)^2\right]. \quad (4)$$

For a money convexity example, consider again BRWA's five-year, 3.2% semiannual coupon bond, settling 15 October 2025, maturing 15 October 2030, and yielding 3.2%—thus, priced at par. We saw previously that annual modified duration and annual convexity for this bond are 4.58676 and 24.23895, respectively.

Suppose the investor has a position in the bond with a par value of USD100 million and the yield-to-maturity increases by 100 bps. We saw earlier that the bond's price declines to 95.53212, so the market value of the position decreases to $95,532,116 (= $100,000,000 × 0.9553212), a decline of $4,467,884.

Estimating the market value change using *MoneyDur* alone results in a larger decline, $4,586,759 (= $458,675,875 × 0.0100), a difference of $118,875.

However, using Equation 3, the *MoneyCon* for this position is $2,423,894,503 (= 24.23895 × $100,000,000), and if we apply Equation 4, which uses both *MoneyDur* and *MoneyCon*, to estimate the market value change, the result is a decline of –$4,465,564, as follows:

$$-\$4,465,564 \approx -(\$458,675,875 \times 0.01) + \left[\frac{1}{2} \times 2,423,894,503 \times (0.01)^2\right].$$

By using both MoneyDur and MoneyCon, the difference in the estimated versus the actual changes in bond position value is only $2,320. Exhibit 5 summarizes these results and shows the estimated and actual changes in position value for a 100 bp decrease in yield.

Exhibit 5:

Using Money Duration and Money Convexity to Estimate Bond Price Changes

Change in Bond Yield	Actual ΔPV^{Full}	Estimated ΔPV^{Full} Using *MoneyDur*	Difference from Actual Change	Estimated ΔPV^{Full} Using *MoneyDur* + *MoneyCon*	Difference from Actual Change
+ 100 BPS	–$4,467,884	–$4,586,759	$118,875	–$4,465,564	–$2,320
– 100 BPS	$4,710,348	$4,586,759	–$123,590	4,707,953	–$2,395

Knowledge Check

bonds settling 11 May 2025 and maturing 11 April 2030. The yield-to-maturity is –0.72%, stated as an effective annual rate on an Act/Act basis. There are five annual periods, and settlement is 30 days into the 365-day year.

1. Calculate the full price of the bond per 100 of par value.

Hide Solution

Solution:

Because Excel's PRICE function does not work for negative yields, the equation for *PV* of a zero-coupon bond must be used. There are five annual periods, settlement is 30 days into the 365-day year, and because 1 + *r* = 1 + (–0.0072) = 0.9928, the full price of the bond is 103.6175 per 100 of par value:

$$PV_0 = \left[\frac{100}{(0.9928)^5}\right] \times (0.9928)^{\frac{30}{365}}$$

PV_0 = 103.617526.

2. Calculate *ApproxModDur* and *ApproxCon* using a 1 bp increase and decrease in the yield-to-maturity.

Hide Solution

Solution:

PV_+ = 103.566215, and PV_- = 103.668868:

$$PV_+ = \left[\frac{100}{(0.9929)^5}\right] \times (0.9929)^{\frac{30}{365}}$$

PV_+ = 103.566215.

$$PV_- = \left[\frac{100}{(0.9927)^5}\right] \times (0.9927)^{\frac{30}{365}}$$

PV_- = 103.668868.

The approximate modified duration is 4.9535:

$$ApproxModDur = \frac{103.668868 - 103.566215}{2 \times (0.0001) \times 103.617526} = 4.9535.$$

The approximate convexity is 29.918:

$$ApproxCon = \frac{103.668868 + 103.566215 - (2 \times 103.617526)}{(0.0001)^2 \times 103.617526} = 29.918.$$

3. Calculate the estimated convexity-adjusted percentage price change resulting from a 100 bp increase in the yield-to-maturity.

Hide Solution

Solution:

The convexity-adjusted percentage price change from a 100 bp increase in yield-to-maturity is estimated as –4.80391%. Modified duration alone estimates the change as –4.9535%. The convexity adjustment adds 14.96 bps.

$$\%\Delta PV^{Full} \approx (-AnnModDur \times \Delta Yield) + \left[\frac{1}{2} \times AnnConvexity \times (\Delta Yield)^2\right].$$

$$\%\Delta PV^{Full} \approx (-4.9535 + 0.01) + \left[\frac{1}{2} \times 29.918 \times (0.01)^2\right]$$

$\%\Delta PV^{Full} \approx -0.048039 = -4.8039\%$.

4. Compare the estimated percentage price change with the actual change, assuming the yield-to-maturity jumps 100 bps, to 0.28%, on the settlement date.

The new full price if the yield-to-maturity jumps from –0.72% to 0.28% on the settlement date is 98.634349:

$$PV^{Full} = \left[\frac{100}{(1.0028)^5}\right] \times (1.0028)^{\frac{30}{365}}$$

PV^{Full} = 98.634349

The actual percentage change in the bond price is –4.80920% (98.634349/103.617526 − 1). The convexity-adjusted estimate is –4.80391%, whereas the estimated change using modified duration alone is –4.95350%.

Question Set

Questions 1–4 relate to the following information.

An investor purchases a €10 million semi-annual 2.95% coupon bond with a yield-to-maturity of 2.95%, settling 30 June 2025 and maturing 30 June 2032.

1. Calculate *ApproxModDur* and *ApproxCon* for the bond using a 1 bp increase and decrease in yield-to-maturity.

   Hide Solution

   Solution:

   PV_+ = PRICE(DATE(2025,6,30),DATE(2032,6,30),0.0295,0.0296,100,2,0) = 99.937.

   PV_- = PRICE(DATE(2025,6,30),DATE(2032,6,30),0.0295,0.0294,100,2,0) = 100.063.

   $$ApproxModDur = \frac{100.063 - 99.937}{2 \times (0.0001) \times 100} = 6.283.$$

   $$ApproxCon = \frac{100.063 + 99.937 - (2 \times 100)}{(0.0001)^2 \times 100} = 44.965.$$

2. Calculate *MoneyDur* and *MoneyCon* for the bond position using the results from Question 1.

   Hide Solution

   Solution:

   MoneyDur ≈ 6.283 × EUR10,000,000 = EUR62,829,180.

   MoneyCon ≈ 44.965 × EUR10,000,000 = EUR449,647,660.

3. Calculate the estimated price change in the investor's position in the bond using *MoneyDur* and *MoneyCon* for a 50 bp decrease in yield-to-maturity.

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   Solution:

   $\Delta PV^{Full} \approx -(\text{EUR}62{,}829{,}180 \times -0.005) + [\tfrac{1}{2} \times \text{EUR}449{,}647{,}660 \times (-0.005)^2]$ = EUR319,767.

4. Calculate the difference between the estimated price change and the actual price change given a 50 bp yield-to-maturity decrease.

   Hide Solution

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PV^{Full} = PRICE(DATE(2025,6,30),DATE(2032,6,30),0.0295,0.0345,100,2,0) = 103.198.

The actual increase in the bond price is 3.1984%:

ΔPV^{Full} = 3.1984% × $10,000,000 = EUR319,840.

The difference between the actual and the estimated price change is EUR73 (= 319,840 – 319,767).

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Discussion

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MF

I'm not sure what I am doing wrong here but I don't get the ApproxCon for Question 1... If I do the exact formula, the numerator brings me to 0 as 100.063+99.9376 = 200 , -2x100 = 0.. then divide that by 0.000001 it gives me 0, not 44.965. Am i missing something here?

Created 3 months ago by Madyson Ferron-McCaffrey 2 replies | Last Activity: 3 months ago

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BN

An investor holds a position in five-year, zero-coupon Federal Republic of Germany bonds settling 11 May 2025 and maturing 11 April 2030. The yield-to-maturity is –0.72%, stated as an effective annual rate on an Act/Act basis. There are five annual periods, and settlement is 30 days into the 365-day year.
Calculate the full price of the bond per 100 of par

In the above Knowledge Check question, there seems to be an inconsistent assumption. From 11 May 2025 to 11 April 2030, there are 4 years and 335 days.(assuming no leap year) . The solution assumes for 5 years and 30 days. If we assume the starting point as 11 April 2025 to get 5 years and 30 days, then the PV comes as 103.74.

Created 9 months ago by Bhuvana Narayanan 0 replies | Last Activity: 9 months ago

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